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DATA MINING

PROJECT TITLE:

**PREDICTING ELECTRICITY CONSUMPTION THROUGH ANALYSIS OF USAGE
PATTERN USING DATA MINING APPROACHES**

GROUP 8

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ABSTRACT

This project focuses on electricity usage forecasting through studying the patterns of usage using data mining techniques. Due to the rising demand for electricity worldwide as a consequence of population growth, urbanisation, and the use of additional electric devices, accurate forecasting is needed for efficient energy management. This research explores the use of data mining techniques such as random forest, neural networks, and linear regression in constructing models of monthly electricity consumption based on household characteristics and lifestyle trends. From a dataset derived from a national smart energy monitoring system, the research models eight predictor variables such as household size, number of appliances, and income level to predict consumption. Pre-processing of the data is carried out in order to handle missing values, and then models are trained and validated using performance measures like mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and R^2 . The findings aim to identify the most important drivers of power usage and the most accurate predictive model, offering implications for energy providers to optimize resource allocation and promote energy efficiency. The finding should be the identification of the best model for accurate prediction of electricity usage and highlighting the main drivers of energy usage.

Keywords

Data Mining; Random Forest; Neural Networks; Cross Validation; Data Partition

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CHAPTER ONE: INTRODUCTION

1.0 Introduction

One of the essential resources that drives contemporary economies and daily living is electricity. Electricity consumption keeps rising due to factors like population growth, urbanisation, and a greater reliance on smart technologies and electrical appliances. More precise and effective techniques for estimating electricity consumption are required as energy demand rises to facilitate load forecasting, efficient energy management, and policymaking.

The volume and complexity of contemporary energy consumption data are frequently too much for traditional forecasting methods to handle. Nonetheless, new developments in data mining techniques offer strong instruments for locating hidden trends and patterns in huge datasets. These methods can be used to examine past data on power use and provide precise predictions about future trends in consumption. This aids consumers, legislators, and energy suppliers in making well-informed choices that improve resource allocation and lessen their negative effects on the environment. To contribute to more effective energy management methods, this project intends to use these cutting-edge data mining techniques to examine patterns of electricity usage and create reliable models for forecasting future demand.

1.1 Problem Statement

The time of day, season, weather, and consumer behaviour are some of the many variables that affect the demand for electricity. Issues including overproduction, underproduction, energy waste, blackouts, and higher operating expenses can result from inaccurate forecasts. Particularly in the context of large data, current forecasting techniques frequently lack the flexibility to manage these dynamic and nonlinear patterns.

Furthermore, utility firms lose out on the chance to optimise supply and minimise energy losses if they don't conduct a thorough examination of patterns of electricity demand. A strong, data-driven strategy is required to forecast electricity usage precisely and offer useful information to consumers and energy suppliers.

1.2 Research Question

1. What are the most influential factors in forecasting electricity consumption patterns?
2. What methods are commonly used to predict electricity usage trends?
3. Which prediction model demonstrates the highest accuracy based on validation results?

1.3 Research Objective

1. To investigate the most dominant factors in predicting monthly electricity usage.
2. To develop a monthly electricity usage prediction model using various data mining techniques.
3. To examine the model evaluation in projecting the accuracy of the prediction model.

1.4 Research Scope

Using data mining techniques on past power usage information, this study aims to anticipate electricity consumption. Only structured data with time-based consumption records will be used in the study; characteristics like the time of day, day of the week, temperature, season, and user type (residential, commercial, etc.) may also be included. Using publicly accessible datasets or simulated data, the research will evaluate how well various data mining methods perform in terms of computational efficiency and prediction accuracy. Real-time energy tracking, hardware-based solutions, and analysis of renewable energy generation are not included in the scope.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

Predicting electricity usage is more crucial today as a result of increased demand and sustainability issues. Data mining helps in this by extracting patterns from large data sets. It starts with data preparation and cleaning (Alsabbagh et al., 2020), then the use of models like regression and decision trees to make predictions of usage in relation to time, weather, and behavior (Zhang et al., 2025; Chen et al., 2025). Neural networks provide deeper pattern analysis and improved accuracy (Bensmaine et al., 2025). However, data validity and authentication remain the backbone of good predictions (Dabaieh et al., 2025).

2.1 The Importance of Electricity Consumption Prediction

Importance of Electricity Consumption Forecasting Accurate forecasting of electricity consumption is important for planning, cost savings, and network integrity. The growing complexity of consumption data, driven by time, weather, and social patterns, renders it challenging with traditional methods. Higher-order methods like MLR, random forest, and neural networks improve the forecast accuracy (Alsabbagh et al., 2020; Gör, 2025). Orange, an interactive data mining environment, simplifies this process through pre-processing functionalities, model training, and testing (Boukhetala & Amirat, 2025).

2.1.1 Application of Data Mining in Electricity Forecasting

Execution of Data Mining Data mining helps to uncover hidden trends in electricity use. Neural networks acquire complex, non-linear relationships (Yin et al., 2024) well, and random forests suppress variance by employing ensemble learning (Rahman et al., 2023). Both suffer from flaws: neural networks need huge data sets and computational power; random forests overfit noisy datasets. Model validation and feature selection improve reliability and performance (Qureshi et al., 2024).

2.1.2 Factors of Electricity Consumption

- **HouseholdSize:** More residents usually mean higher usage (Ugbehe et al., 2025).
- **NumAppliances:** More devices increase electricity needs (Qureshi et al., 2024).
- **HasAirConditioner:** A/C units significantly raise usage (Rahman et al., 2023).
- **AvgDailyUsageHours:** Longer usage times mean higher consumption (Easwaramoorthy et al., 2020).
- **IncomeLevel:** Higher incomes may imply more appliances and larger homes (Boukhetala & Amirat, 2025).
- **DwellingType:** Detached houses often use more energy due to greater exposure to outdoor temperatures (Qureshi et al., 2024).
- **Region:** Climate and energy policies influence consumption (Boukhetala & Amirat, 2025).

2.2 Electricity Consumption Prediction Using Various Techniques

Electrical consumption forecasting plays an important role in modern energy management to set up sustainability, operational efficiency, and strategic management. Due to increased electrical consumption due to urbanization, technological progress, and population growth, the use of effective and precise forecasting techniques becomes necessary. In this chapter, an introduction to some major forecasting techniques commonly utilized to forecast electrical consumption is given, such as multiple linear regression (MLR), neural networks, random forest, decision trees, and support vector machines (SVM).

2.2.1 Multiple Linear Regression (MLR)

MLR is a technique that explains the relationship between consumption and a number of variables. It is easy to comprehend and useful in analyzing prominent factors like temperature and lighting. Studies carried out by UTHM and UTeM confirm that MLR explains over 80% variance in energy (Aises Journal, 2023; Sustainable Energy Research, 2025).

2.2.2 Neural Networks

Neural networks capture intricate relationships in data and have been found better than classical models to deal with varying daily usage (Mehmood et al., 2024; Yin et al., 2024). Its limitations are high resource usage and more time-consuming training.

2.2.3 Random Forest

Random forest, an ensemble algorithm, constructs a set of decision trees to enhance prediction power and avoid overfitting. It's helpful in establishing the most important factors influencing usage (Boukhetala & Amirat, 2025)

2.3 Gaps and Opportunities for Improvement

While MLR is easy to use, it lacks the capability of modeling non-linear data. This is overcome by neural networks and random forests but not leveraged since they are complicated (Gör, 2025). In this research, the gap is bridged through the use of Orange to evaluate models based on metrics like RMSE, MSE, and R^2 (Alsabbagh et al., 2020). Orange's feature-ranking helps identify principal factors, improving model precision and interpretability, especially for features like Dwelling Type, HasAirConditioner, and AvgDailyUsageHours (Ugbehe et al., 2025; Boukhetala & Amirat, 2025).

CHAPTER THREE: METHODOLOGY

3.0 Introduction

This chapter includes a detailed explanation regarding the research process on how the prediction model of monthly electricity usage is constructed. To start with, descriptive analysis contains the discussion about the dataset used for this study. Then, it leads to the research process which contains five phases, starting with Data Selection as Phase 1 and ends with Expected Output as Phase 5.

3.1 Descriptive Analysis

The dataset is extracted from a national smart energy monitoring system. It contains valuable information from 200 households which was collected as part of a nationwide smart grid pilot program. The objective of this dataset is to predict the monthly electricity consumption in kWh, based on household characteristics and lifestyle habits. Predictive models can be built to help energy providers in identifying high-usage patterns and propose efficiency measures. There are 8 variables in this dataset which are, HouseholdSize, NumAppliances, HasAirConditoner, AvgDailyUsageHours, IncomeLevel, DwellingType, Region, and finally,

MonthlyElectricityUsage, which is the target variable. By using Feature Statistics in Orange software, the distribution of all 8 variables including the target variable can be plotted.

Furthermore, to achieve the objective of this study, the Rank feature in Orange software will be used to determine the most prominent variables in the relation of predicting the monthly electricity usage in kWh. The primary evaluation method used for the prediction model is Mean Squared Error, which is the average of the squared differences between actual and predicted values. The lower the value, the better the model in predicting the monthly electricity usage. Other evaluation methods such as RMSE, MAE, MAPE, and R^2 , are used to further solidify the consistency of the model.

3.2 Research Process

Five phases of the research process will be discussed. It starts with Data Selection as Phase 1, and it is followed by Data Pre-Processing, Model Development, Model Evaluation and Expected Output as Phase 2, 3, 4, and 5, respectively as shown in Figure 1 in Appendix.

3.2.1 Phase 1: Data Selection

As mentioned in the descriptive analysis, the dataset is extracted from a national smart energy monitoring system, and it consists of 8 attributes. All variables but one are chosen as the input for the analysis. As indicated, 7 attributes act as predictor variables and one attribute acts as a target variable. Table 1 in Appendix includes the list of variables with its descriptions for each variable respectively.

3.2.2 Phase 2: Data Pre-Processing

The dataset extracted contains missing values in the region column. Data cleaning process is done using the Impute feature in the Orange Software to replace the missing values with the most frequent data.

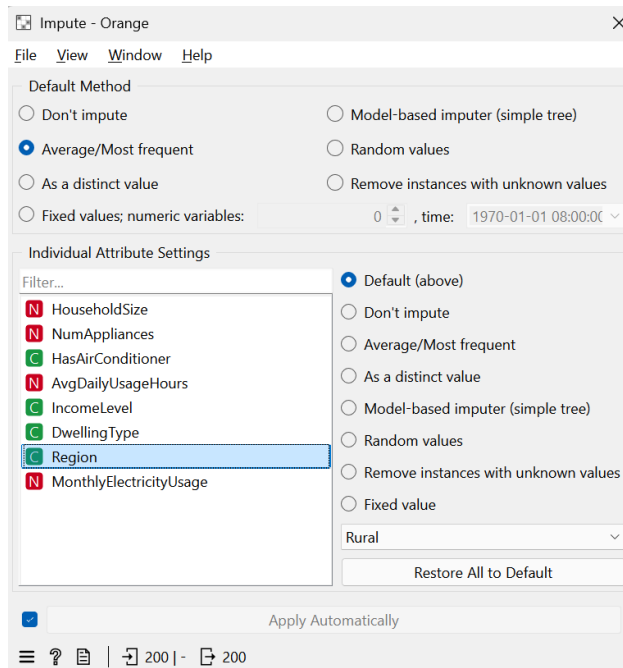


Figure 2: Data cleaning process using Impute Feature in Orange.

No variables were removed throughout data pre-processing. In addition, the data is partitioned according to three data partitions which are 60:40, 70:30 and 80:20 whereas the first group of data will act as training data and the second group of data will act as testing data. For instance, 70% out of 200 instances will be used as training data while the remaining 30% will be used as testing data.

3.2.3 Phase 3: Model Development

Upon completing phase 2, a prediction model will be developed in the Orange software through three techniques in predicting the monthly electricity usage which are Random Forest, Linear Regression and Neural Network as shown in Figure 3.

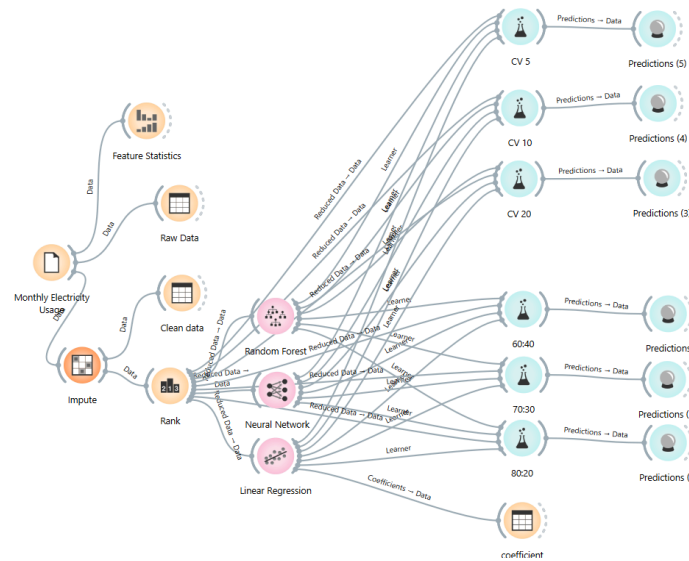


Figure 3: Model Development

- **Random Forest**

Random Tree is a set of decision trees. Each tree is developed from a bootstrap sample from the training data. When developing individual trees, a random subset of attributes is drawn which gives its name, random forest, from which the best attribute for the split is selected. The final model is based on the majority vote from individually developed trees in the forest. Figure 4 shows the settings used for the Random Forest model. It includes 10 random trees, while the model does not split subsets smaller than 5 instances. There is no limit to the number of attributes considered at each split, and no limit on the depth of each individual tree.

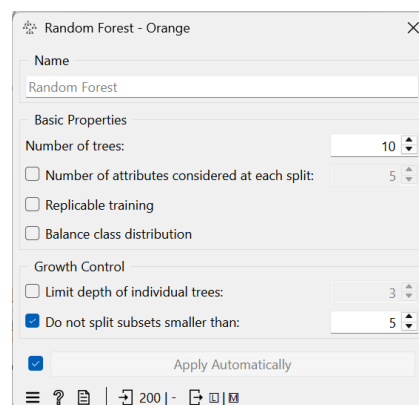


Figure 4: Settings for Random Forest

- **Neural Network**

Neural networks is a process of connecting data using nodes. These nodes are an inputs layer, hidden layer with weights, and an output layer. Data is mapped through supervised learning, like how the human brain is interconnected with neurons. This model can be programmed to give threshold values to determine a model's accuracy. Figure 5 shows the settings for the Neural Network model. The number of neurons in the hidden layers is set to 100, the activation is set as ReLu, and the solver is set as Adam as it is more widely used for most cases. The maximal number of iterations is 200, and each training is replicable.

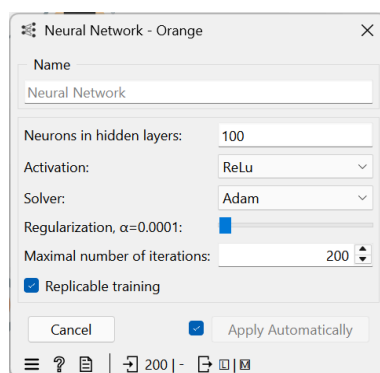


Figure 5: Settings For Neural Network

- **Linear Regression**

Linear regression is a technique that predicts the value of unknown data by using other related and known data values. It mathematically models the unknown or dependent variable and the known or independent variable as a linear equation. Figure 6 shows the settings for Linear Regression.

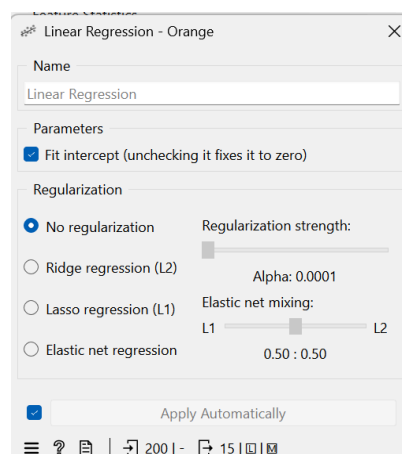


Figure 6: Settings for Linear Regression

3.2.4 Phase 4: Model Evaluation

The prediction model will be evaluated by performance metrics for regression models specifically for this dataset. In Orange, the performance metrics for the regression model consist of Mean Square Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). All except R-squared describe the better model as the one with a lower value. In contrast, for R-squared, it has a maximum value of 1 where the higher the value means a better model. A negative value deems the model terrible for prediction.

Each method will be tested using Random Sampling by dividing into different data partitions. Based on data partition in Phase 2, the prediction models are categorized as Table 4 in Appendix, with the intention to examine different evaluation results in choosing the best monthly electricity usage prediction model.

On the other hand, each method will also be tested using Cross Validation which includes a number of folds of 5, 10, and 20. The evaluation method is similar to the Random Sampling test. Refer to Table 5 in Appendix.

To further identify whether some variables are as important as the other, the data will be split into 3 groups of Top 3 variables, Top 5 variables, and Top 7 which includes all variables. Random Forest will be tested by changing the number of trees to 10, 100, and 200. Neural Network will be tested by changing the neurons in hidden layers to 100, 200, (100,100), and (200,200). Linear Regression will be tested with Ridge, Lasso, and no regularisation throughout the testing while the regularisation strength alpha remains as 0.0001.

3.2.5 Phase 5: Expected Output

Expected Output concludes the evaluation from previous phases with a predicted result of monthly electricity usage. Besides, the prediction model will be generated from the model development and the expected findings of this study are to discover the most prominent factors of diabetes using the Rank feature. Followed by the best prediction model chosen based on the performance metrics. The expected best prediction model is the Linear Regression technique, while the top 3 input variables to predict monthly electricity usage are DwellingType, HasAirConditioner, and AvgDailyUsageHour.

3.3 Summary

It is important to predict the monthly electricity usage to better understand consumer's needs and how electricity providers can propose efficiency measures by identifying high-usage patterns. The prediction can be done through the chosen data mining techniques which are Random Forest, Neural Network, and Linear Regression. These methods are chosen since they are suitable to handle numerical and continuous data. Orange software makes the most efficient use of such information of each technique to successfully predict the monthly electricity usage. On top of that, various evaluation methods will be used such as Mean Square Error (MSE), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2), in investigating the credibility and the accuracy of the prediction model. Lastly, the most accurate model will be chosen as the best technique to predict monthly electricity usage.

CHAPTER FOUR: RESULTS AND DISCUSSION

4.0 Introduction

This chapter will analyse the evaluation results generated by Orange according to each method and settings used. The best results obtained will be thoroughly discussed, and finally, the overall best method and its settings is revealed at the end of this chapter.

4.1 Linear Regression

- **Cross Validation**

The results of different regularisation for cross validation is surprisingly similar in all evaluation tests. This suggests that the changes in regularisation is insignificant when the regularisation strength, alpha, is only set to 0.0001. A higher alpha could see different results for each regularisation. That being said, the best results for Linear Regression are obtained when Rank is Top 5, with a number of folds of 10. *Table 3: Best Evaluation Results for Linear Regression Cross Validation.*

Regression	Rank	Cross Validation (Folds)	Evaluation Result			
			MSE	MAE	MAPE	R ²
No Regularisation	Top 5	10	35.493	3.671	0.019	0.981
Ridge Regression	Top 5	10	35.493	3.671	0.019	0.981
Lasso Regression	Top 5	10	35.493	3.671	0.019	0.981

- **Data Partition**

Similar to cross validation, the results from different regularisations are similar. The best results for MSE and R² are better than Cross Validation, and interestingly, the MSE, and MAE for Lasso Regression are slightly better compared to other regularisations, at Rank Top 5 and Data Partition of 70/30.

Regression	Rank	Data Partition	Evaluation Result			
			MSE	MAE	MAPE	R ²
Lasso Regression	Top 5	70/30	32.675	3.764	0.019	0.983

Table 3: Best Evaluation Results for Linear Regression Data Partition.

4.2 Neural Networks

- **Cross Validation**

The number of neurons in hidden layer of 100 and 200 showed terrible evaluation results with MSE above 30,000 and negative R² values no matter the rank and number of folds. However, for (100,100) and (200, 200), a more acceptable result is observed, where the best results come from (200, 200) with Rank Top 5 and number of folds of 10.

Neurons in hidden layer	Rank	Cross Validation (Folds)	Evaluation Result			
			MSE	MAE	MAPE	R ²
200, 200	Top 5	10	142.493	9.24	0.048	0.922

Table 3: Best Evaluation Results for Neural Networks Cross Validation.

- **Data Partition**

The overall evaluation results for Data Partition is better than Cross Validation. Similarly, only (100, 100) and (200, 200) showed acceptable results, with (200, 200) generating the best evaluation results at Rank Top 5 and Data Partition of 80/20.

Neurons in hidden layer	Rank	Data Partition	Evaluation Result			
			MSE	MAE	MAPE	R2
200, 200	Top 5	80/20	110.994	8.564	0.043	0.939

Table 3: Best Evaluation Results for Neural Networks Data Partition.

4.3 Random Forest

- **Cross Validation**

The usual misconception for random forest is that when the number of trees increases, the accuracy increases. This is proven as not always true in Cross Validation, where the best result is obtained when the number of trees is 100, at Top 5 Rank, and number of folds of 20.

Number of Trees	Rank	Cross Validation (Folds)	Evaluation Result			
			MSE	MAE	MAPE	R2
100	Top 5	20	171.119	9.89	0.05	0.906

Table 3: Best Evaluation Results for Neural Networks Cross Validation.

- **Data Partition**

For Data Partition however, the best result comes from the higher number of trees of 200, at Top 5 Rank, and the Data Partition of 80/20.

Number of Trees	Rank	Data Partition	Evaluation Result			
			MSE	MAE	MAPE	R2
200	Top 5	80/20	209.864	11.065	0.055	0.884

Table 3: Best Evaluation Results for Neural Networks Data Partition.

4.4 Overall Result and Discussions

Upon filtering out the best evaluation results for all three different tests, the overall best result can be easily identified to be from Linear Regression. Comparing between the two methods within Linear Regression shows that Data Partitioning, specifically 70:30, produces the best results. Types of regression for Linear Regression highlights that Lasso regression shows a better result than Ridge and No regularisation, though the difference is minute. The reasoning behind the improvement can be concluded by the randomness of the Orange software when a test is conducted. That said, the Linear Regression test could come out as equal for all three regularisations when the test is repeated in future testing.

The inaccuracy or high errors of Neural Networks test can be explained by the lack of normalised and standardised data set. A noisy data set that has sudden fluctuation of larger values can deteriorate the testing process. Factors that may have influenced Random Forest could be the small number of trees. A higher number of trees could increase the accuracy but will take extra time to run and test. Overall, completely different evaluation results could be produced using a standardised data set or with a larger number of trees, however, due to time constraints, only a limited number of variables was tested, while Linear Regression showed better clear cut results with the variables tested.

It is obvious which Rank showed the best results for all tests, that is from Top 5 features. A top 3 features can be too biased as the test is only based on 3 attributes. Key information might be excluded, which in turn hurts prediction accuracy. Top 7 features could include too many irrelevant and noisy features. This introduces unwanted noise and can lead to overfitting or reduced generalization. Overall, the top 5 features are balanced and provide just enough attributes for the models to provide the best accuracy.

Cross Validation is usually more preferred over Data Partition, but the overall results show that Data Partition is more accurate. This is highly due to a lucky split in partitioning the data set. Cross Validation having a slightly lower accuracy could be due to a truer generalisation performance.

With the knowledge of Linear Regression being the best model to test and predict electricity consumption through analysis of usage patterns, companies such as electricity providers or electrical appliance manufacturers can use Linear Regression models to analyse the

electricity consumption of a consumer. This analysis can inspire new electricity saving plans or alternate electrical products that saves electricity cost or consumptions entirely.

5.0 CONCLUSION

Analysis of usage patterns to predict electricity consumption can be done efficiently and accurately using Linear Regression model in Orange software. Other models like Neural Networks and Random Forest are viable options, but will take more effort and time to obtain the perfect settings or data environment, such as data normalisation, to match or get close to the accuracy provided from the Linear Regression model. The top 3 attributes that contribute to the testing are DwellingType, HasAirConditioner, and AvgDailyUsageHours.

Electricity providers like Tenaga Nasional Berhad can utilise this analysis to provide sufficient electricity supply to consumers according to their usage patterns. Besides, consumer consultations or FAQs regarding their electricity bills can now be answered and explained accordingly using analysis of usage patterns of consumers. Higher electricity bills often relate to higher electricity consumption where certain usage patterns may emerge and relationships can be drawn using Linear Regression models.

Companies aiming to provide energy saving electrical appliances such as air conditioners, can identify specifically which products contribute to a higher electricity consumption by looking at the usage patterns of each consumer. Understanding consumers will surely benefit both the customer by improving their lifestyle, and the companies to provide better products catering to energy saving for general consumers.

As mentioned before, time constraint is the main limitation of this study, followed by a relatively small data set. A more thorough and detailed study could be carried out in the future by testing out other factors such as, conducting Linear Regression models using a higher regularisation strength. This could help identify which regularisation is better among the Linear Regression models. Better results for Neural Networks could also be obtained by standardising the data set before running it through the model. The same can be said for Random Forest, which could be experimented with a higher number of trees. Finally, the analysis should be conducted with a larger data set to generate a more accurate result, with less bias, and more than just 7 attributes.

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APPENDIX

No.	Prediction Technique	Authors & Year
1.	Back-propagation Neural Network	Song et al., 2019
2.	Deep Learning	Qureshi et al., 2024
3.	Data Mining	Easwaramoorthy et al., 2020
4.	Data Mining	Rahman et al., 2023
5.	Machine Learning	Gör, 2025
6.	Data Mining	Dabaieh et al., 2025

Table 1: Selected Electricity Consumption Prediction Studies by Technique

Authors/ Factor	Househo -ld Size	Num Applianc -es	Has Air Conditio -ner	Average Daily Usage Hours	Income Level	Dwelling Type	Region
Ugbehe -et al., 2025	/	/					
Qureshi -et al., 2024		/			/		
Rahman -et al., 2023			/				

Veerappa mpalaya- m Easwara- moorthy -at al.,2020				/			
Easwara- moorthy -et al., 2020			/		/	/	/

Table 2: Factors of Electricity Consumption

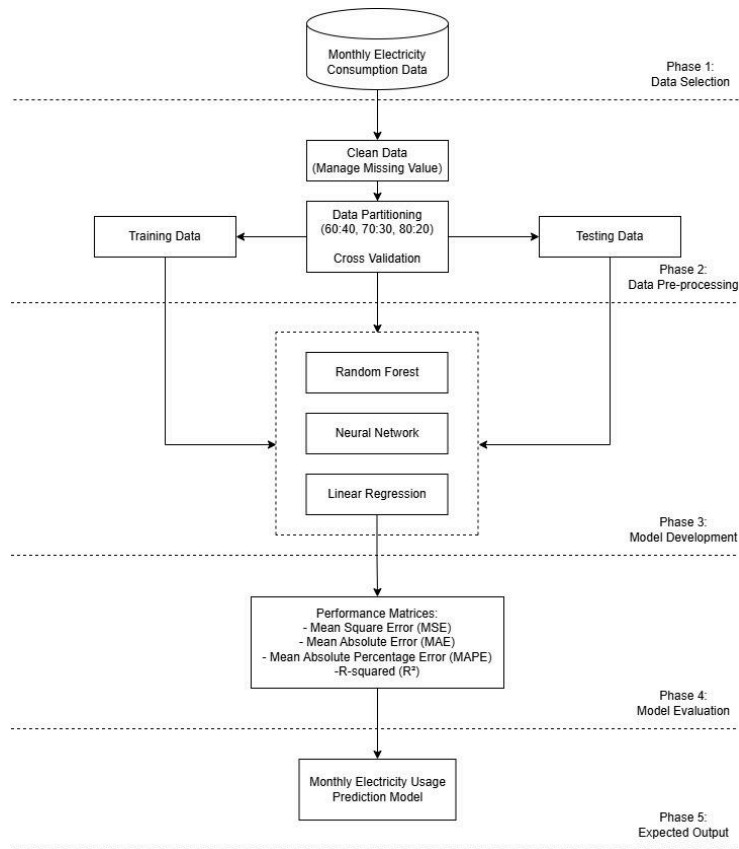


Figure 1: Research Process of Monthly Electricity Consumption Prediction.

No.	Variable	Role	Type	Values	Description
1.	HouseholdSize	Feature	Numeric	-	Number of people living in the household
2.	NumAppliances	Feature	Numeric	-	Number of electrical appliances owned
3.	HasAirConditioner	Feature	Categorical	Yes, No	Whether the household uses air conditioning
4.	AvgDailyUsageHours	Feature	Numeric	-	Average hours of electricity usage per day
5.	IncomeLevel	Feature	Categorical	High, Low, Medium	Household income level
6.	DwellingType	Feature	Categorical	Apartment, Detached, Terrace	Type of house
7.	Region	Feature	Categorical	Rural, Suburban, Urban	Area where the household is located
8.	MonthlyElectricityUsage	Target	Numeric	-	Monthly electricity usage in kilowatt-hours (kWh)

Table 3: Data Description.

Methods	Data Partition	Evaluation Results			
		MSE	MAE	MAPE	R ²
Random Forest	60/40	-	-	-	-
	70/30	-	-	-	-
	80/20	-	-	-	-
Neural Network	60/40	-	-	-	-
	70/30	-	-	-	-
	80/20	-	-	-	-
Linear Regression	60/40	-	-	-	-
	70/30	-	-	-	-
	80/20	-	-	-	-

Table 4: Model Evaluation According to Data Partition (Performance Parameters)

Methods	Number of Folds	Evaluation Results			
		MSE	MAE	MAPE	R ²
Random Forest	5	-	-	-	-
	10	-	-	-	-
	20	-	-	-	-
Neural Network	5	-	-	-	-
	10	-	-	-	-
	20	-	-	-	-
Linear Regression	5	-	-	-	-
	10	-	-	-	-
	20	-	-	-	-

Table 5: Model Evaluation According to Number of Folds (Performance Parameters)